Quaternions

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Roadmap

- Number theory over "exotic" domains to prove two nontrivial theorems (Fermat's Christmas theorem, Lagrange's four-square theorem)
- A first taste of ring theory

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First definitions

- The complex numbers $\mathbb{C}$ are made by adding $\sqrt{-1}$ to $\mathbb{R}$.
- The quaternions $\mathbb{H}$ are made by adding *three* different $\sqrt{-1}$s to $\mathbb{R}$.

Definition

A quaternion is a number of the form

$$a = a_1 + a_2 i + a_3 j + a_4 k$$

for reals a_1, a_2, a_3, a_4 and the "square roots of -1" i, j, and k. Addition is done in the obvious way. Multiplication is distributive over addition and associative, and

$$i^2 = j^2 = k^2 = -1 \qquad ij = -ji = k \qquad jk = -kj = i \qquad ki = -ik = j.$$

First definitions

- Multiplication is not in general commutative, but the real quaternions commute with every element. So $\lambda h = h\lambda$ for all $\lambda \in \mathbb{R}$ and $h \in \mathbb{H}$.
- The *conjugate* of $a = a_1 + a_2 i + a_3 j + a_4 k$ is $a^* = a_1 - a_2 i - a_3 j - a_4 k$.
- The *norm* of a is $|a| = \sqrt{a_1^2 + a_2^2 + a_3^2 + a_4^2} = \sqrt{aa^*} = \sqrt{a^* a}$.
- We have $|ab| = |a|\,|b|$.
- We have $(ab)^* = b^* a^*$. In general, this is not equal to $a^* b^*$.
- The multiplicative inverse of a is $a^{-1} = \frac{a^*}{|a|^2}$.
- Usually we don't write $\frac{a}{b}$, unless one of a or b is real, since multiplication isn't commutative.

History

Every morning in the early part of October 1843, on my coming down to breakfast, your brother William Edwin and yourself used to ask me: "Well, Papa, can you multiply triples?" Whereto I was always obliged to reply, with a sad shake of the head, "No, I can only add and subtract them."

—A letter from Hamilton to his son Archibald

And here there dawned on me the notion that we must admit, in some sense, a fourth dimension of space for the purpose of calculating with triples... An electric circuit seemed to close, and a spark flashed forth.

—A letter from Hamilton to his friend and colleague John Graves, inventor of the octonions, an eight-dimensional number system

History

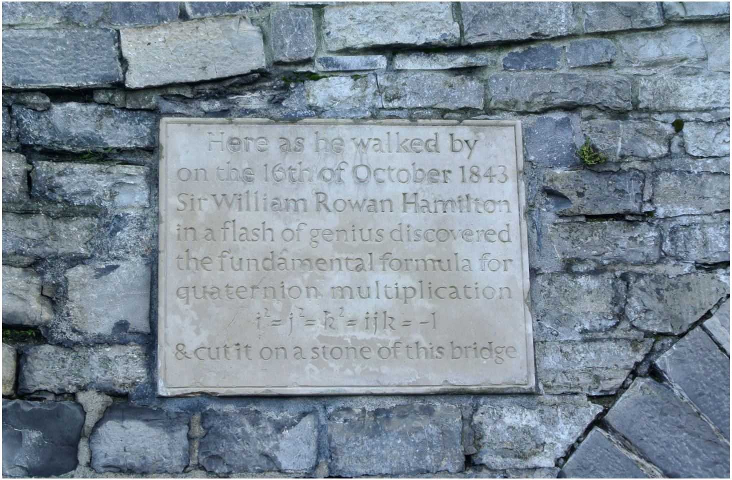

Figure: A plaque on the Brougham Bridge commemorating the discovery of quaternions. Observe the identity $i^2 = j^2 = k^2 = ijk = -1$.

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Euclid's lemma

The goal of this section is to prove the following intuitive lemma.

Lemma (Euclid's lemma)

If p is a prime and $p \mid mn$, then $p \mid m$ or $p \mid n$.

(Actually, in ring theory, this is how you *define* a prime. Don't worry about that for now. Here a prime number is one that's divisible only by 1 and itself.)

Euclidean algorithm setup

- For $d, n \in \mathbb{Z}$, we say d *divides* n, written $d \mid n$, if $n = dk$ for some $k \in \mathbb{Z}$.
- For $p \in \mathbb{Z}$, we say p is *prime* if $d \mid p$ implies $|d| = 1$ or $|d| = p$.

The following three facts will allow us to perform the Euclidean algorithm.

If $d \mid m, n$, then $d \mid m - kn$ for $k \in \mathbb{Z}$.

Let $m = ad$ and $n = bd$, then $m - kn = ad - kbd = (a - kb)d$.

For all $x \in \mathbb{R}$, there is some $k \in \mathbb{Z}$ such that $|x - k| < 1$.

Easy.

For all $m, n \in \mathbb{Z}$ and $n \neq 0$, there is some $k \in \mathbb{Z}$ such that $|m - kn| < |n|$.

Let $x = \frac{m}{n}$, there is some $k \in \mathbb{Z}$ such that $\left|\frac{m}{n} - k\right| < 1$. Multiplying by $|n|$ gives $|m - kn| < |n|$.

The Euclidean algorithm

- If $d \mid m, n$, then $d \mid m - kn$ for $k \in \mathbb{Z}$.
- For all $m, n \in \mathbb{Z}$ and $n \neq 0$, there is some $k \in \mathbb{Z}$ such that $|m - kn| < |n|$.

Let's do the Euclidean algorithm:

- Start with integers m and n (WLOG $|m| \geq |n|$).
- Let $(m_0, n_0) = (m, n)$.
- For $i \geq 1$, let $(m_{i+1}, n_{i+1}) = (n_i, m_i - k_i n_i)$ where k_i is chosen such that $|m_i - k_i n_i| < |n|$.
- Let $\gcd(m, n)$ be the value of m_i when n_i first reaches 0.
- There are integers a and b such that $\gcd(m, n) = am + bn$.

A concrete example

Example with $(m, n) = (30, 21)$:

$$\begin{aligned}
(m_0, n_0) &= (30, 21) = (m, n) \\
(m_1, n_1) &= (21, 9) = (n, m - n) \\
(m_2, n_2) &= (9, 3) = (m - n, -2m + 3n) \\
(m_3, n_3) &= (3, 0) = (-2m + 3n, 7m - 10n).
\end{aligned}$$

Note that:

- At each step, m_i and n_i are linear combinations of m and n.
- In particular, $\gcd(m, n) = m_3 = 3$ is a linear combination of m and n. That is, for any m and n, there exist integers a and b such that $am + bn = \gcd(m, n)$. This is known as Bézout's identity.
- Also, $\gcd(m, n)$ divides each m_i and n_i.

Try the cases $(m, n) = (24, 20)$ and $(m, n) = (42, -30)$.

Euclid's lemma

Lemma (Euclid's lemma)

If p is a prime and $p \mid mn$, then $p \mid m$ or $p \mid n$.

Proof.

Suppose $p \mid mn$ and $p \nmid m$. Then $|\gcd(p, m)| = 1$, so by Bézout's identity there are integers a and b such that $ap + bm = 1$. So $apn + bmn = n$, but since $p \mid apn + bmn$, $p \mid n$. $\square$

Now let's do the same thing over the Gaussian integers!

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Some definitions

The Gaussian integers are the set

$$\mathbb{Z}[i] = \{a + bi \mid a, b \in \mathbb{Z}\} \subseteq \mathbb{C}.$$

Observe that

- For all $n \in \mathbb{Z}[i]$, $|n|^2$ is an integer.
- The Gaussian integers are closed under addition and multiplication.

We set up the exact same definitions:

- For $d, n \in \mathbb{Z}[i]$, we say d *divides* n, written $d \mid n$, if $n = dk$ for $k \in \mathbb{Z}[i]$.
- For $p \in \mathbb{Z}[i]$, we say p is *prime* if $d \mid p$ implies $|d| = 1$ or $|d| = |p|$.

We will prove Euclid's lemma again:

Lemma (Euclid's lemma)

Let $p, m, n \in \mathbb{Z}[i]$ with p prime. If $p \mid mn$, then $p \mid m$ or $p \mid n$.

The same three claims

If $d \mid m, n$, then $d \mid m - kn$ for $k \in \mathbb{Z}[i]$.

Let $m = ad$ and $n = bd$, then $m - kn = ad - kbd = (a - kb)d$.

For all $x \in \mathbb{C}$, there is some $k \in \mathbb{Z}[i]$ such that $|x - k| < 1$.

Let $x = x_1 + x_2 i$ and $k = k_1 + k_2 i$, we may force $|x_1 - k_1| \leq \frac{1}{2}$ and $|x_2 - k_2| \leq \frac{1}{2}$, then $|x - k| = \sqrt{(x_1 - k_1)^2 + (x_2 - k_2)^2} \leq \frac{\sqrt{2}}{2} < 1$.

For all $m, n \in \mathbb{Z}[i]$ and $n \neq 0$, there is some $k \in \mathbb{Z}[i]$ such that $|m - kn| < |n|$.

Let $x = \frac{m}{n}$, there is some $k \in \mathbb{Z}[i]$ such that $\left|\frac{m}{n} - k\right| < 1$. Multiplying by $|n|$ gives $|m - kn| < |n|$.

The same Euclidean algorithm

- Start with $m, n \in \mathbb{Z}[i]$ (WLOG $|m| \geq |n|$).
- Let $(m_0, n_0) = (m, n)$.
- For $i \geq 1$, let $(m_{i+1}, n_{i+1}) = (n_i, m_i - k_i n_i)$ where k_i is chosen such that $|m_i - k_i n_i| < |n|$.
- Let $\gcd(m, n)$ be the value of m_i when n_i first reaches 0.
- There are $a, b \in \mathbb{Z}[i]$ such that $\gcd(m, n) = am + bn$.

A concrete example

Example with $(m, n) = (-6 + 7i, 3 + 4i)$:

$$\begin{aligned}
(m_0, n_0) &= (-6 + 7i, 3 + 4i) = (m, n) \\
(m_1, n_1) &= (3 + 4i, -6 + 7i - 2i(3 + 4i)) \\
&= (3 + 4i, 2 + i) = (n, m - 2in) \\
(m_2, n_2) &= (2 + i, 3 + 4i - (2 + i)(2 + i)) \\
&= (2 + i, 0) = (m - 2in, (-2 - i)m + (-1 + 4i)n)
\end{aligned}$$

Again, note that:

- At each step, m_i and n_i are linear combinations of m and n.
- In particular, $\gcd(m, n) = m_2 = 2 + i$ is a linear combination of m and n.
- Also, $\gcd(m, n)$ divides each m_i and n_i.

Euclid lemma 2: electric boogaloo

Lemma (Euclid's lemma)

Let $p, m, n \in \mathbb{Z}[i]$ with p prime. If $p \mid mn$, then $p \mid m$ or $p \mid n$.

Proof.

Suppose $p \mid mn$ and $p \nmid m$. Then $|\gcd(p, m)| = 1$, so by Bézout's identity there are Gaussian integers a and b such that $ap + bm = \gcd(p, m)$, or equivalently, multiplying by $\overline{\gcd(p, m)}$, there are Gaussian integers a' and b' such that $a'p + b'm = 1$. So $apn + bmn = n$, but since $p \mid apn + bmn$, $p \mid n$. $\square$

Fermat's Christmas theorem

This is enough to extract the following:

Theorem (Fermat's Christmas theorem)

Let $p \in \mathbb{Z}$ be a prime. Then $p = x^2 + y^2$ for some $x, y \in \mathbb{Z}$ iff $p = 2$ or $p \equiv 1 \pmod 4$.

Note that $2 = 1^2 + 1^2$ and $3 \pmod 4$ primes fail modulo 4. So we just need to show each $1 \pmod 4$ prime is the sum of two squares.

Lemma

If $p \equiv 1 \pmod 4$, then $p \mid 1 + a^2$ for some $a \in \mathbb{Z}$.

- Write $p = 4k + 1$ and consider the polynomial $x^{4k} - 1 = (x^{2k} - 1)(x^{2k} + 1)$ modulo p.
- This has $p - 1 = 4k$ roots modulo p by FLT, so $x^{2k} - 1$ and $x^{2k} + 1$ each have exactly $2k$ roots by counting degrees.
- So $x^{2k} + 1$ has some root modulo p, so $p \mid (x^k)^2 + 1$, done.

Fermat's Christmas theorem

Lemma

Any $1 \pmod 4$ *prime* p *is not prime in the Gaussian integers.*

- If p was prime, $p \mid 1 + a^2 = (1 + ai)(1 - ai)$ for some $a \in \mathbb{Z}$, so $p \mid 1 + ai$ or $p \mid 1 - ai$ by Euclid's lemma.
- But neither $\frac{1}{p} + \frac{a}{p}i$ nor $\frac{1}{p} - \frac{a}{p}i$ are Gaussian integers, done.

Now here's the main proof:

- Since p is not irreducible in $\mathbb{Z}[i]$, write $p = mn$ with $|m|, |n| \neq 1, |p|$.
- Squaring and taking the norm, $|m|^2 |n|^2 = p^2$, but since $|m|^2, |n|^2 \in \mathbb{Z}$, $|m|^2 = |n|^2 = p$.
- If $m = x + yi$, then $|m|^2 = x^2 + y^2 = p$, as desired.

Now let's do the same thing over the Hurwitz integers!

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The Hurwitz integers

Definition

The Hurwitz quaternions H are the set of quaternions whose coefficients are either all integers or all half-integers. In particular, Hurwitz integers cannot have a mix of integer and half-integer coefficients.

Observe that

- For all $h \in H$, $|h|^2$ is an integer.
- The Hurwitz integers are closed under addition and multiplication.

The same definitions

- For $h, n \in H$, we say d *right divides* h, written $d \mid_{\rightarrow} h$, if $h = ad$ for some $a \in H$. We say d *left divides* h, written $d \mid_{\leftarrow} h$, if $h = db$ for some $b \in H$.[1]
- A Hurwitz integer p is called *prime* if $d \mid_{\rightarrow} p$ implies $|d| = 1$ or $d = |p|$.

We require two notions of divisibility because multiplication is not commutative.

We will prove the following. Looks familiar?

Lemma (Weak Euclid's lemma)

Let $p \in H$ be a real Hurwitz prime. Then if $p \mid mn$, $p \mid m$ or $p \mid n$.

[1] I just made up the notation $\mid_{\rightarrow}$ and $\mid_{\leftarrow}$ for right and left divisibility. I'm not sure if any standard notation exists.

The same three claims, again

If $d \mid_{\rightarrow} m, n$, then $d \mid_{\rightarrow} m - hn$ for $h \in H$.

Let $m = ad$ and $n = bd$, then $m - hn = ad - hbd = (a - hb)d$.

For all $x \in \mathbb{H}$, there is some $h \in H$ such that $|x - k| < 1$.

Write $x = x_1 + x_2 i + x_3 j + x_4 k$ and $h = h_1 + h_2 i + h_3 j + h_4 k$. We may force $|x_1 - h_1| \leq \frac{1}{4}$ and $|x_i - h_i| \leq \frac{1}{2}$ for $i = 2, 3, 4$. Then $|x - h| = \sqrt{(x_1 - h_1)^2 + \sum_{i=2}^{4}(x_i - h_i)^2} \leq \sqrt{\left(\frac{1}{4}\right)^2 + 3\left(\frac{1}{2}\right)^2} < 1$, as desired.

For all $m, n \in H$ and $n \neq 0$, there is some $h \in H$ such that $|m - hn| < |n|$.

Let $x = mn^{-1}$, then there is some $h \in H$ such that $\left|mn^{-1} - h\right| < 1$. Right multiplying by $|n|$, $|m - hn| < |n|$, as desired.

The Euclidean algorithm, again

- Start with $m, n \in H$ (WLOG $|m| \geq |n|$).
- Let $(m_0, n_0) = (m, n)$.
- For $i \geq 1$, let $(m_{i+1}, n_{i+1}) = (n_i, m_i - k_i n_i)$ where k_i is chosen such that $|m_i - k_i n_i| < |n|$.
- Let $\gcd(m, n)$ be the value of m_i when n_i first reaches 0.
- There are $a, b \in H$ such that $\gcd(m, n) = am + bn$.

Weak Euclid's lemma

Lemma (Weak Euclid's lemma)

Let $p \in H$ be a real Hurwitz prime. Then if $p \mid mn$, $p \mid m$ or $p \mid n$.

Proof.

Suppose $p \mid mn$ and $p \nmid m$. Then $|\gcd(p, m)| = 1$, so by Bézout's identity there are Hurwitz integers a and b such that $ap + bm = \gcd(p, m)$, or equivalently, left multiplying by $\gcd(p, m)^*$, there are Hurwitz integers a' and b' such that $a'p + b'm = 1$. So $apn + bmn = n$, but since $p \mid apn + bmn$, $p \mid n$. $\square$

Lagrange's four-square theorem

Now this is enough to extract the following.

Theorem (Lagrange's four-square theorem)

Every positive integer is the sum of four squares.

- Observe that 1 works.
- We must show that for every positive integer a, there is some quaternion with integer coefficients m such that $|m|^2 = a$.
- Since $|mn| = |m|\,|n|$, it's enough to show the result for all primes.
- Actually, since $|1+i| = \sqrt{2}$, it's enough to show the result for all odd primes.

Lagrange's four-square theorem

Lemma

For odd primes p, $p \mid 1 + a^2 + b^2$ for some integers a and b.

- There are $\frac{p+1}{2}$ perfect squares modulo p.
- So there are $\frac{p+1}{2}$ elements of the form $-1 - a^2$ modulo p.
- So by pigeonhole, since $\frac{p+1}{2} + \frac{p+1}{2} > p$, there must be some perfect square b^2 that is of the form $-1 - a^2$ modulo p.
- So $-1 - a^2 \equiv b^2 \pmod{p}$ or $p \mid 1 + a^2 + b^2$.

Lemma

Any odd prime p is not prime in the Hurwitz integers.

- If p was prime, since $p \mid 1 + a^2 + b^2 = (1 + ai + bj)(1 - ai - bj)$ for some $a, b \in \mathbb{Z}$, $p \mid 1 + ai + bj$ or $p \mid 1 - ai - bj$ by weak Euclid's lemma.
- But neither $\frac{1}{p} + \frac{a}{p}i + \frac{b}{p}j$ nor $\frac{1}{p} - \frac{a}{p}i - \frac{b}{p}j$ are Hurwitz integers.

Lagrange's four-square theorem

Here's the main proof:

- Since p is not irreducible in H, write $p = mn$ with $|m|, |n| \neq 1, p$.
- Squaring and taking the norm, $|m|^2 |n|^2 = p^2$, but since $|m|^2, |n|^2 \in \mathbb{Z}$, $|m|^2 = |n|^2 = p$.
- If m has integer coefficients, we are done. Otherwise, pick $\epsilon = \frac{1}{2}(\pm 1 \pm i \pm j \pm k)$ such that $m' := m + \epsilon$ has even integer coefficients.
- But then $p = mm^* = (m' - \epsilon)(m'^* - \epsilon^*) = (m' - \epsilon)\epsilon^* \epsilon (m^* - \epsilon^*) = (m'\epsilon^* - 1)(\epsilon m'^* - 1)$.
- Since m' has even integer coefficients, $m'\epsilon^* - 1$ and $\epsilon m'^* - 1$ have integer coefficients.
- But $|m'\epsilon^* - 1| = |\epsilon m'^* - 1| = \sqrt{p}$, which concludes.

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The cross product and quaternions

Consider the vectors $i, j, k \in \mathbb{R}^3$ with $i = (1, 0, 0)$, $j = (0, 1, 0)$, and $k = (0, 0, 1)$. The cross product gives

$$\begin{aligned} i \times j &= -j \times i = k \\ j \times k &= -k \times j = i \\ k \times i &= -i \times k = j. \end{aligned}$$

Look familiar? Remember the quaternion multiplication identities

$$ij = -ji = k \qquad jk = -kj = i \qquad ki = -ik = j.$$

The cross product and quaternions

Consider the quaternions with no real part $x = x_1 i + x_2 j + x_3 k$ and $y = y_1 i + y_2 j + y_3 k$. Then if $xy = z_1 i + z_2 j + z_3 k$, then

$$\begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} \times \begin{pmatrix} y_1 \\ y_2 \\ y_3 \end{pmatrix} = \begin{pmatrix} z_1 \\ z_2 \\ z_3 \end{pmatrix}.$$

Observe that the *right-hand rule* implies there are two ways to define the cross product. Similarly, there are two ways to define quaternion multiplication, one may define $ij = -k$ instead.

Octonions

If with your alchemy you can make three pounds of gold, why should you stop there?

—John T. Graves (inventor of the octonions) in a letter to Hamilton

- An *eight*-dimensional number system invented by Graves with *seven* square roots of -1
- Not even associative, but alternative: $x(xy) = (xx)y$ and $(yx)x = y(xx)$.

The octonion multiplication table

e_ie_j	e_0	e_1	e_2	e_3	e_4	e_5	e_6	e_7
e_0	e_0	e_1	e_2	e_3	e_4	e_5	e_6	e_7
e_1	e_1	$-e_0$	e_3	$-e_2$	e_5	$-e_4$	$-e_7$	e_6
e_2	e_2	$-e_3$	$-e_0$	e_1	e_6	e_7	$-e_4$	$-e_5$
e_3	e_3	e_2	$-e_1$	$-e_0$	e_7	$-e_6$	e_5	$-e_4$
e_4	e_4	$-e_5$	$-e_6$	$-e_7$	$-e_0$	e_1	e_2	e_3
e_5	e_5	e_4	$-e_7$	e_6	$-e_1$	$-e_0$	$-e_3$	e_2
e_6	e_6	e_7	e_4	$-e_5$	$-e_2$	e_3	$-e_0$	$-e_1$
e_7	e_7	$-e_6$	e_5	e_4	$-e_3$	$-e_2$	e_1	$-e_0$

Figure: The octonion multiplication table with $e_0 = 1$ and the imaginary units $e_1, \ldots, e_7$. Screenshot from Wikipedia to avoid formatting a table in LaTeX.

Relevant quote on octonions

God had no hand in the creation of this abhorrence. The fact that this [number system] exists proves that God is either impotent to alter His universe or ignorant to the horrors taking place in His kingdom. [The octonions are] more than [a number system]. It is a physical declaration of mankind's contempt for the natural order. It is hubris manifest.

—`r/copypasta`

The seven-dimensional cross product

- Related to the octonions the same way the three-dimensional cross product is related to the quaternions.
- There are 480 ways to define the seven-dimensional cross product, compared to only 2 with the three-dimensional cross product.
- Similarly to the three-dimensional case, $|a \times b| = |a|\,|b| \sin\theta$, where θ is the angle between a and b, $a, b \perp a \times b$, and $a \times b = -b \times a$.
- The cross product *only exists* in three and seven dimensions.